Engelking General Topology Benchmark

Natural language statement	Domain	GroundTruth

Natural language statement: Definition. A topological space X is called a realcompact space if X is a Tychonoff space and there is no Tychonoff space X which satisfies: (RC1) there exists a homeomorphic embedding r: X -> X such that r(X) != closure(r(X)) = X; (RC2) for every continuous real-valued function f: X -> R there exists a continuous function f: X -> R such that fr = f. 3.11.16. Theorem. For every Tychonoff space X there exists exactly one (up to a homeomorphism) realcompact space nuX which satisfies: (i) there exists a homeomorphic embedding nu: X -> nuX such that closure(nu(X)) = nuX; (ii) for every continuous real-valued function f: X -> R there exists a continuous function f^nu: nuX -> R such that f^nunu = f. The space nuX also satisfies: (iii) for every continuous mapping f: X -> Y of X to a realcompact space Y there exists a continuous mapping f^nu: nuX -> Y such that f^nunu = f.

Domain: Topology

GroundTruth:

```
def IsRealcompact (X : Type u) [tX : TopologicalSpace X] : Prop :=
  T35Space X /\ not exists (Y : Type (u + 1)) (tY : TopologicalSpace Y),
    @T35Space Y tY /\ exists r : X -> Y,
      @IsEmbedding X Y tX tY r /\ range r != closure (range r) /\
        closure (range r) = univ /\
          forall f : X -> R, Continuous f ->
            exists g : Y -> R, @Continuous Y R tY _ g /\ g comp r = f

theorem engelking_3_11_16_hewitt_realcompactification
    {X : Type u} [TopologicalSpace X] [T35Space X] :
    exists (Y : Type u) (tY : TopologicalSpace Y) (nu : X -> Y),
      @IsRealcompact Y tY /\ @IsEmbedding X Y _ tY nu /\ closure (range nu)
      = univ /\
        (forall f : X -> R, Continuous f ->
          exists g : Y -> R, @Continuous Y R tY _ g /\ g comp nu = f) /\
        (forall (Z : Type u) (tZ : TopologicalSpace Z), @IsRealcompact Z tZ
        ->
          forall f : X -> Z, @Continuous X Z _ tZ f ->
            exists g : Y -> Z, @Continuous Y Z tY tZ g /\ g comp nu = f) /\
        forall (Z : Type u) (tZ : TopologicalSpace Z) (zeta : X -> Z),
          @IsRealcompact Z tZ -> @IsEmbedding X Z _ tZ zeta -> closure
        (range zeta) = univ ->
          (forall f : X -> R, Continuous f ->
            exists g : Z -> R, @Continuous Z R tZ _ g /\ g comp zeta = f) ->
          exists h : @Homeomorph Y Z tY tZ, h comp nu = zeta := by
  sorry
```

Natural language statement: Definition. A topological space X is called countably paracompact if X is a Hausdorff space and every countable open cover of X has a locally finite open refinement. 5.2.8. Theorem. A topological space X is normal and countably paracompact if and only if the Cartesian product X x I of X and the closed unit interval I is normal.

Domain: Topology

GroundTruth:

```
def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}
    (U : iota -> Set X) : Prop :=
  (forall i, IsOpen (U i)) /\ union i, U i = univ

def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set
X)
    (U : iota -> Set X) : Prop :=
  forall j, exists i, V j subset U i

def IsCountablyParacompact (X : Type u) [TopologicalSpace X] : Prop :=
  T2Space X /\
    forall (iota : Type v) (_ : Countable iota) (U : iota -> Set X),
    IsOpenCover U ->
      exists (kappa : Type v) (V : kappa -> Set X), IsOpenCover V /\
      Refines V U /\ LocallyFinite V

theorem engelking_5_2_8_normal_product_interval
    {X : Type u} [TopologicalSpace X] [T1Space X] :
    (NormalSpace X /\ IsCountablyParacompact X) <->
      NormalSpace (X  x  Set.Icc (0 : R) 1) := by
  sorry
```

Natural language statement: Definitions. A family {A_s}_{sinS} is star-finite (star-countable) if for every s0 in S the set {s in S : A_s inter A_{s0} != empty} is finite (countable). A topological space X is called strongly paracompact if X is a Hausdorff space and every open cover of X has a star-finite open refinement. 5.3.10. Theorem. For every regular space X the following conditions are equivalent: (i) X is strongly paracompact; (ii) every open cover of X has a closed refinement which is both locally finite and star-finite; (iii) every open cover of X has a closed refinement which is both locally finite and star-countable; (iv) every open cover of X has a star-countable open refinement.

Domain: Topology

GroundTruth:

```
def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}
    (U : iota -> Set X) : Prop :=
  (forall i, IsOpen (U i)) /\ union i, U i = univ

def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set
X)
    (U : iota -> Set X) : Prop :=
  forall j, exists i, V j subset U i

def IsClosedCover {X : Type u} [TopologicalSpace X] {iota : Type v}
    (F : iota -> Set X) : Prop :=
  (forall i, IsClosed (F i)) /\ union i, F i = univ

def IsStarFiniteFamily {X : Type u} {iota : Type v} (A : iota -> Set X) :
Prop :=
  forall i, {j | (A i inter A j).Nonempty}.Finite

def IsStarCountableFamily {X : Type u} {iota : Type v} (A : iota -> Set X)
: Prop :=
  forall i, {j | (A i inter A j).Nonempty}.Countable

def IsStronglyParacompact (X : Type u) [TopologicalSpace X] : Prop :=
```

<table>
<tr><th>Natural language statement</th><th>Domain</th><th>GroundTruth</th></tr>
<tr><td></td><td></td><td>T2Space X /\
forall (iota : Type v) (U : iota -> Set X), IsOpenCover U ->
exists (kappa : Type v) (V : kappa -> Set X),
IsOpenCover V /\ Refines V U /\ IsStarFiniteFamily V

theorem engelking_5_3_10_strong_paracompactness
{X : Type u} [TopologicalSpace X] [RegularSpace X] [T1Space X] :
let closedLocallyFiniteStarFinite :=
forall (iota : Type v) (U : iota -> Set X), IsOpenCover U ->
exists (kappa : Type v) (F : kappa -> Set X),
IsClosedCover F /\ Refines F U /\ LocallyFinite F /\
IsStarFiniteFamily F
let closedLocallyFiniteStarCountable :=
forall (iota : Type v) (U : iota -> Set X), IsOpenCover U ->
exists (kappa : Type v) (F : kappa -> Set X),
IsClosedCover F /\ Refines F U /\ LocallyFinite F /\
IsStarCountableFamily F
let starCountableOpen := forall (iota : Type v) (U : iota -> Set X),
IsOpenCover U ->
exists (kappa : Type v) (V : kappa -> Set X),
IsOpenCover V /\ Refines V U /\ IsStarCountableFamily V
List.TFAE [IsStronglyParacompact X, closedLocallyFiniteStarFinite,
closedLocallyFiniteStarCountable, starCountableOpen] := by
sorry</td></tr>
<tr><td>4.4.1. The Stone Theorem. Every open cover of a metrizable space has an open refinement which is both locally finite and sigma-discrete.</td><td>Topology</td><td>def IsDiscreteFamily {X : Type u} [TopologicalSpace X] {iota : Type v}
(U : iota -> Set X) : Prop :=
forall x : X, exists V in nhds x, {i : iota | (V inter U
i).Nonempty}.Subsingleton

def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}
(U : iota -> Set X) : Prop :=
(forall i, IsOpen (U i)) /\ union i, U i = univ

def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set
X)
(U : iota -> Set X) : Prop :=
forall j, exists i, V j subset U i

theorem engelking_4_4_1_stone_open_refinement
{X : Type u} [TopologicalSpace X] [MetrizableSpace X] :
forall (iota : Type v) (U : iota -> Set X), IsOpenCover U ->
exists (kappa : Type w) (V : kappa -> Set X) (level : kappa -> N),
IsOpenCover V /\ Refines V U /\ LocallyFinite V /\
forall n, IsDiscreteFamily fun j : {j : kappa // level j = n} =>
V j := by
sorry</td></tr>
<tr><td>4.4.7. The Nagata-Smirnov Metrization Theorem. A topological space is metrizable if and only if it is regular and has a sigma-locally finite base.</td><td>Topology</td><td>def HasSigmaLocallyFiniteBase (X : Type u) [TopologicalSpace X] : Prop :=
exists (iota : N -> Type v) (B : forall n, iota n -> Set X),
IsTopologicalBasis {V : Set X | exists n i, B n i = V} /\
forall n, LocallyFinite (B n)

theorem engelking_4_4_7_nagata_smirnov_metrization
{X : Type u} [TopologicalSpace X] [T1Space X] :
MetrizableSpace X <-> RegularSpace X /\ HasSigmaLocallyFiniteBase X :=
by
sorry</td></tr>
<tr><td>4.4.8. The Bing Metrization Theorem. A topological space is metrizable if and only if it is regular and has a sigma-discrete base.</td><td>Topology</td><td>def IsDiscreteFamily {X : Type u} [TopologicalSpace X] {iota : Type v}
(U : iota -> Set X) : Prop :=
forall x : X, exists V in nhds x, {i : iota | (V inter U
i).Nonempty}.Subsingleton

def HasSigmaDiscreteBase (X : Type u) [TopologicalSpace X] : Prop :=
exists (iota : N -> Type v) (B : forall n, iota n -> Set X),
IsTopologicalBasis {V : Set X | exists n i, B n i = V} /\
forall n, IsDiscreteFamily (B n)

theorem engelking_4_4_8_bing_metrization
{X : Type u} [TopologicalSpace X] [T1Space X] :</td></tr>
</table>

Natural language statement	Domain	GroundTruth
		`MetrizableSpace X <-> RegularSpace X /\ HasSigmaDiscreteBase X := by` `sorry`

5.1.9. Theorem. For every T1-space X the following conditions are equivalent: (i) the space X is paracompact; (ii) every open cover of the space X has a locally finite partition of unity subordinated to it; (iii) every open cover of X has a partition of unity subordinated to it.	Topology	`def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}`
		`(U : iota -> Set X) : Prop :=`
		`(forall i, IsOpen (U i)) /\ union i, U i = univ`
		
		`theorem engelking_5_1_9_paracompact_partition_of_unity`
		`{X : Type u} [TopologicalSpace X] [T1Space X] :`
		`let locallyFinitePartition := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U ->`
		`exists rho : PartitionOfUnity iota X, rho.IsSubordinate U`
		`let partition := forall (iota : Type v) (U : iota -> Set X), IsOpenCover U ->`
		`exists rho : iota -> C(X, R), 0 <= rho /\ (forall x, HasSum (fun i => rho i x) 1) /\`
		`forall i, tsupport (rho i) subset U i`
		`List.TFAE [ParacompactSpace X, locallyFinitePartition, partition] := by`
		`sorry`

5.1.38. The Tamano Theorem. For every Tychonoff space X the following conditions are equivalent: (i) the space X is paracompact; (ii) for every compactification cX of the space X the Cartesian product X x cX is normal; (iii) the Cartesian product X x betaX is normal; (iv) there exists a compactification cX of X such that X x cX is normal.	Topology	`def IsCompactification {X : Type u} {K : Type v} [TopologicalSpace X]`
		`[TopologicalSpace K] (e : X -> K) : Prop :=`
		`IsEmbedding e /\ DenseRange e /\ IsCompact (univ : Set K) /\ T2Space K`
		
		`theorem engelking_5_1_38_tamano_theorem`
		`{X : Type u} [tX : TopologicalSpace X] [T35Space X] :`
		`let everyCompactification := forall (K : Type v) (tK : TopologicalSpace K) (e : X -> K),`
		`@IsCompactification X K tX tK e ->`
		`@NormalSpace (X x K) (tX.induced Prod.fst tK.induced Prod.snd)`
		`/\ T1Space (X x K)`
		`let someCompactification := exists (K : Type v) (tK : TopologicalSpace K) (e : X -> K),`
		`@IsCompactification X K tX tK e /\`
		`@NormalSpace (X x K) (tX.induced Prod.fst tK.induced Prod.snd)`
		`/\ T1Space (X x K)`
		`List.TFAE [ParacompactSpace X, everyCompactification,`
		`NormalSpace (X x StoneCech X) /\ T1Space (X x StoneCech X),`
		`someCompactification] := by`
		`sorry`

7.2.1. The Countable Sum Theorem. If a normal space X has a countable closed cover {F_j} such that dim F_j <= n for j = 1,2,..., then dim X <= n.	Topology	`def IsOpenCover {X : Type u} [TopologicalSpace X] {iota : Type v}`
		`(U : iota -> Set X) : Prop :=`
		`(forall i, IsOpen (U i)) /\ union i, U i = univ`
		
		`def Refines {X : Type u} {iota : Type v} {kappa : Type w} (V : kappa -> Set X)`
		`(U : iota -> Set X) : Prop :=`
		`forall j, exists i, V j subset U i`
		
		`def CoverOrderLE {X : Type u} {iota : Type v} (U : iota -> Set X) (n : N) : Prop :=`
		`forall x : X, forall s : Finset iota, (forall i in s, x in U i) -> s.card <= n + 1`
		
		`def CoveringDimensionLE (X : Type u) [TopologicalSpace X] (n : N) : Prop :=`
		`forall (m : N) (U : Fin m -> Set X), IsOpenCover U ->`
		`exists (k : N) (V : Fin k -> Set X),`
		`IsOpenCover V /\ Refines V U /\ CoverOrderLE V n`
		
		`theorem engelking_7_2_1_countable_sum_theorem`
		`{X : Type u} [TopologicalSpace X] [NormalSpace X] {n : N}`
		`(hcover : exists F : N -> Set X, (forall j, IsClosed (F j)) /\ union j, F j = univ /\`
		`forall j, CoveringDimensionLE (F j) n) :`
		`CoveringDimensionLE X n := by`
		`sorry`

8.4.13. The Smirnov Theorem. Assigning to every compactification cX of a Tychonoff space X the proximity	Topology	`def IsCompactification {X : Type u} {K : Type v} [TopologicalSpace X]` `[TopologicalSpace K] (e : X -> K) : Prop :=`

Natural language statement	Domain	GroundTruth

delta(c) establishes a one-to-one correspondence between compactifications of X and proximities on X.		`IsEmbedding e /\ DenseRange e /\ IsCompact (univ : Set K) /\ T2Space K`
		
		`def EquivalentCompactifications {X : Type u} {K L : Type v} [TopologicalSpace X]`
		`[TopologicalSpace K] [TopologicalSpace L] (e : X -> K) (f : X -> L) : Prop :=`
		`exists h : K equivt L, h comp e = f`
		
		`structure Proximity (X : Type u) [TopologicalSpace X] where`
		`close : Set X -> Set X -> Prop`
		`empty_left : forall A, not close empty A`
		`intersects : forall A B, (A inter B).Nonempty -> close A B`
		`symmetric : forall A B, close A B <-> close B A`
		`union_left : forall A B C, close (A union B) C <-> close A C \/ close B C`
		`strong : forall A B, not close A B -> exists E : Set X, not close A E /\ not close E^c B`
		`closure_eq : forall A : Set X, closure A = {x | close {x} A}`
		
		`def IsAssignedProximity {X : Type u} {K : Type v} [TopologicalSpace X]`
		`[TopologicalSpace K] (e : X -> K) (p : Proximity X) : Prop :=`
		`forall A B : Set X,`
		`p.close A B <-> (closure (e ’’ A) inter closure (e ’’ B)).Nonempty`
		
		`theorem engelking_8_4_13_smirnov_proximity_compactification`
		`{X : Type u} [TopologicalSpace X] [T35Space X] :`
		`(forall (K : Type v) (tK : TopologicalSpace K) (e : X -> K),`
		`@IsCompactification X K _ tK e ->`
		`exists p : Proximity X, @IsAssignedProximity X K _ tK e p) /\`
		`(forall p : Proximity X, exists (K : Type v) (_ : TopologicalSpace K) (e : X -> K),`
		`IsCompactification e /\ IsAssignedProximity e p) /\`
		`forall (K L : Type v) (_ : TopologicalSpace K) (_ : TopologicalSpace L)`
		`(e : X -> K) (f : X -> L) (p : Proximity X),`
		`IsCompactification e -> IsCompactification f ->`
		`IsAssignedProximity e p -> IsAssignedProximity f p ->`
		`EquivalentCompactifications e f := by`
		`sorry`